

Ranjot Sandhu

Software Engineer • AI/ML Engineer • Full-Stack Developer

Seeking office, hybrid, and remote software engineering roles

ranjotsandhu.com 

[/00ranjotsandhu](https://00ranjotsandhu) 

[/HydralsProgramming](https://HydralsProgramming) 

00ranjotsandhu@gmail.com 

269-213-0348 

PROFESSIONAL SUMMARY

Computer Science graduate (minor in UX Design) with experience across software engineering, AI/ML, site reliability, and full-stack product development through TD Bank (via Tata Consultancy Services), Reality AI Lab, and Wilfrid Laurier University. Reduced manual incident-detection time by more than 50% and launched an esports platform that reached 40 paid subscribers in its first full month. Strong in Python, Java, JavaScript, React, Next.js, SQL, Docker, unit testing (pytest), CI/CD, and cloud deployment, with a UX foundation and experience carrying products through the full software development lifecycle.

EXPERIENCE

Freelance Software Developer | Blue Moon Development

January 2026 - Present

- Operate a development studio delivering production websites, e-commerce, full-stack apps, and automation.
- Led engagements from discovery through UX design, development, testing, deployment, and client support.
- Delivered live commercial products including an e-commerce storefront and a revenue-generating esports platform.
- Deployed and maintained production applications on Vercel and Netlify with streamlined release workflows.
- Managed client communication, proposals, and timelines while ensuring clients retained code and domain ownership.

Software Engineer Intern | Tata Consultancy Services (Client: TD Bank)

June 2025 - August 2025

- Developed monitoring scripts and SRE tooling that cut manual incident-detection time by over 50%.
- Built and maintained Datadog dashboards for production service health, triage, and operational visibility.
- Delivered performance testing, technical reports, sprint updates, and handoff docs in an agile environment.
- Collaborated cross-functionally with software and infrastructure teams on production reliability improvements.
- Ran performance and load testing with JMeter and Dynatrace to identify and resolve system bottlenecks.

Student Research Project Developer | Wilfrid Laurier University

September 2025 - Present

- Designed predictive ML models for faculty-led research, reaching ~85% accuracy via feature engineering.
- Built Python, Pandas, and NumPy pipelines to clean, validate, and structure datasets for model training.
- Translated model results into research-ready conclusions through technical documentation and faculty presentations.
- Built an arm-model research system to identify behavioral patterns and generate actionable insights for the research team.
- Delivered a multi-window Tkinter and Matplotlib GUI with live training telemetry and kinematics visualization.

Software Engineer Intern | Reality AI Lab

February 2025 - May 2025

- Developed and validated AI training workflow components using Python, TensorFlow, PyTorch, Pandas, and Scikit-learn.
- Performed structured QA, regression testing, bug tracking, and output validation for AI-integrated product features.
- Designed Figma wireframes and interface components that connected model capabilities to clearer user-facing workflows.
- Worked across product and engineering tasks to improve training workflow quality, output consistency, and release readiness.
- Collaborated with a small remote team using Git version control across both product and technical decisions.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL, C, Swift, R, HTML, CSS

AI / Data: TensorFlow, PyTorch, Scikit-learn, Pandas, NumPy, Matplotlib, Reinforcement Learning (PPO, SAC, A2C), Machine Learning, Data Visualization

Full Stack: React, Next.js, Node.js, SwiftUI, REST APIs, RESTful API Design, Supabase, PostgreSQL, Gradio, Stripe, OAuth, Authentication & Authorization

Reliability / Tools: Datadog, Dynatrace, JMeter, Postman, Git, Version Control, Figma, Xcode, Tkinter

Deployment / Testing: Docker, Vercel, Netlify, Render, pytest, CI/CD, Agile/Scrum, Unit Testing, Regression Testing, QA

EDUCATION

Wilfrid Laurier University Graduated August 2026

Bachelor of Computer Science, Minor in UX Design | Waterloo, Ontario, Canada

ADDITIONAL EXPERIENCE

Student Proctor & Teaching Assistant | Wilfrid Laurier University January 2024 - August 2026

- Supported exam setup, student check-in, academic integrity, and technical troubleshooting for assessments.
- Coordinated with faculty on course logistics, instructional materials, grading, and exam-day technical requirements.

PROJECTS

All or Nothing Esports | *Next.js, TypeScript, Supabase, Product Growth* allornothing.gg

- Built and launched a multi-game esports platform with ladders, tournaments, subscriptions, and coaching.
- Reached 40 paid subscribers and \$240 MRR in month one; facilitated 38 sessions and \$950 marketplace volume.
- Integrated Discord OAuth authentication/authorization, Supabase, Stripe subscription and payment processing, and admin tools.
- Own product direction, community operations, monetization experiments, deployment, and ongoing feature delivery.

Bookshelf | *Swift, SwiftUI, Supabase, PostgreSQL*

- Built a multi-tenant iOS app for clerks to manage lottery inventory and end-of-shift accounting.
- Designed access control and data security with PostgreSQL Row Level Security across 16 tables; built BLE, HID, and camera barcode workflows.
- Implemented shift lifecycle functions, book swaps, automated sales reconciliation, and per-book inventory tracking.
- Used Supabase authentication and database functions to secure privileged inventory operations serverlessly.

MarketSignal Intelligence Lab | *Python, Gradio, Scikit-learn, Pandas* [Live demo](#)

- Built a market-research workbench comparing Random Forest, Gradient Boosting, and Ridge Regression.
- Engineered time-ordered features, held-out evaluation, model-driver analysis, data visualization, and backtesting.
- Implemented SMA9/SMA21, improved the Gradio workflow, and expanded multi-stock dataset reliability.
- Designed chronological train/test evaluation to reduce look-ahead leakage and expose weak performance.

Arm Model Research System | *Python, R, NumPy, Reinforcement Learning*

- Trained PPO and SAC reinforcement-learning agents to control a robotic-arm simulation for daily-living tasks.
- Designed reward functions and policy logic that achieved more than 90% policy confidence against defined task benchmarks.
- Developed the system as a foundation for future assistive and healthcare exoskeleton research.
- Built data-processing and visualization workflows in Python, R, NumPy, Tkinter, and Matplotlib.

Eileen's English Toffee | *Responsive Web Design, E-commerce, Client Delivery* [Live site](#)

- Designed and delivered a responsive storefront presenting products, brand story, and purchase pathways.
- Translated a small-business brief into an accessible, deployment-ready commercial website with streamlined customer navigation.
- Implemented catalogue, product-detail, cart, owner-access, contact, event, and email-capture experiences.